SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, ima models, and image processing techn and solve real-world Computer Visio***		
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Section / Batch:	**Slot A, 2025**	**Date:**	**14/10/2026**

Code of Conduct

I S.SRIJITHA 192471024 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: srijitha

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text. **Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.**

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Answers:

Q1. Improving nighttime CCTV images before object detection

A CCTV system in a shopping mall may capture clear images during the day but produce dark, noisy images at night because there is less available light and the camera must amplify the signal. That amplification can make sensor noise more visible, while low contrast can hide details such as faces, clothing, and objects. The goal of image preprocessing is to improve useful visual information before object detection, while avoiding transformations that create misleading details. Processing should be consistent across frames so that the detection system can interpret changes in the scene reliably.

The first step is **image acquisition and inspection**. The system can examine each frame's brightness and noise level and, if available, use camera metadata such as exposure time and gain. Images should be represented in a suitable format, such as RGB or a luminance–chroma space. In a luminance–chroma space, brightness can be adjusted largely independently of color, helping preserve the scene's color information. If the image is extremely dark, the system may also use camera-side improvements such as infrared illumination or a longer exposure, provided that the exposure does not create excessive motion blur.

Next, apply **noise removal**. A median filter can reduce isolated bright or dark pixels while preserving many object boundaries. A Gaussian filter is useful for reducing fine, approximately random noise, but it blurs edges as well as noise. When preserving people's outlines and small details is important, an edge-preserving filter, such as a bilateral filter, can smooth areas with similar brightness while retaining stronger boundaries. Denoising should be moderate: overly strong filtering can erase small features, such as a narrow object, facial detail, or a thin edge that the detector needs.

After denoising, perform **contrast and brightness enhancement**. A simple brightness adjustment may make the scene more visible, but increasing all pixel values uniformly can wash out bright areas such as signs or lamps. Contrast stretching can expand a limited range of pixel intensities, making darker and lighter regions more distinct. Histogram equalization can improve global contrast, although it may over-amplify noise or create an unnatural appearance. For scenes with bright and dark regions in the same frame, **adaptive histogram equalization**, often used in a contrast-limited form, can enhance local detail while limiting excessive contrast amplification. The same settings should be used carefully over a video sequence; frame-by-frame changes that are too aggressive can make brightness flicker.

Sharpening can be applied after denoising and contrast enhancement to make edges clearer. A mild unsharp-mask operation enhances details by adding a controlled amount of high-frequency information back to the image. Sharpening should not be performed strongly on a noisy image, because it can make noise look like real texture or edges. It also cannot recover detail that was never captured, such as information lost through severe underexposure or motion blur. If blur is a recurring problem, the camera's exposure and lighting should be addressed rather than relying on sharpening alone.

The recommended sequence before object detection is therefore: **capture and assess the frame; convert to a**

suitable color representation; denoise; correct brightness and enhance contrast; apply mild sharpening; then supply the processed frame to the object detector. Optional steps include resizing to the detector's required input dimensions and normalizing pixel values. The detector should be tested on both original and processed nighttime footage to verify that preprocessing improves recognition without increasing false detections. Performance can be measured using object-detection measures such as precision, recall, and localization accuracy, alongside visual checks for detail preservation. Processing should be applied consistently, since an enhancement that improves one scene but obscures another may reduce overall reliability.

Q2. Improving camera-captured document images

A camera-based document-scanning system must convert a photographed page into an image that is clear, aligned, and suitable for reading or optical character recognition. Shadows, sensor noise, uneven lighting, perspective distortion, and blurred text all affect the result. The processing pipeline should first make the page geometry and illumination more consistent, then distinguish text from the background, and finally divide the page into meaningful regions such as lines, words, or text blocks.

The system begins with **image acquisition and page detection**. The camera should capture the entire page with as little motion as possible, and the system should identify the page boundary. If the page is tilted or viewed at an angle, the detected corners can be used for geometric correction, often called perspective rectification or deskewing. This correction is important because subsequent thresholding and segmentation work best when text lines are approximately horizontal and the page is consistently oriented.

Filtering addresses noise and small unwanted variations. A mild Gaussian filter can suppress fine sensor noise, while a median filter can remove isolated specks without blurring edges as much. For documents, filtering must be restrained because letters contain thin strokes and small gaps. Excessive smoothing can merge characters or remove punctuation. If the image has shadows or uneven illumination, filtering alone is not enough: those problems are broad changes in brightness, not simply fine-scale noise. A background estimate can be used to correct slow illumination variations, for example by dividing or subtracting a smoothed version of the image, with suitable safeguards against changing the text itself.

Contrast enhancement makes the difference between ink and paper more apparent. A contrast stretch can be useful when the whole image has a narrow range of brightness. Local contrast enhancement can help when one part of the page is darker than another, but it must be limited to prevent background texture and compression noise from becoming prominent. A useful objective is a reasonably uniform light page background and clearly visible dark strokes, not merely a visually dramatic image.

Thresholding converts the grayscale image into a binary or near-binary representation of foreground text and background paper. A global threshold uses one intensity cutoff for the entire page and can work well when lighting is uniform and the paper is clean. In photographs with shadows or gradients, adaptive thresholding calculates a local cutoff for different parts of the image and is generally more robust. Thresholding should be performed after illumination correction and appropriate denoising where possible. If applied too early, a shadow may be classified as a dark region, or faint text may disappear.

Edge detection identifies strong intensity transitions that often correspond to character strokes, page

boundaries, tables, or diagrams. Edge maps can help locate the page boundary and support geometric correction. They can also help distinguish text strokes from broad illumination changes. However, edges alone do not reliably identify text: paper texture, lines, and shadows can also generate edges, and broken or blurred strokes may not form complete contours. Edge detection is therefore best treated as a supporting operation rather than a replacement for thresholding or region analysis.

Segmentation divides the cleaned image into useful units. At a coarse level, the system can separate the page from the background. It can then locate text blocks, columns, tables, figures, and other regions. Connected-component analysis on a binary image may group pixels into character-like components, while line-projection methods can help identify text lines. More advanced methods can use layout and spacing information to separate paragraphs or columns. The segmentation method should account for connected handwriting, touching characters, and multi-column pages rather than assuming every document contains isolated printed letters.

A practical pipeline is: **capture the page; detect its outline; correct perspective and rotation; denoise lightly; correct uneven illumination; enhance contrast; threshold the image using a global or adaptive method; use edge information where useful for boundary and layout detection; then segment the page into text and content regions**. The output can then be passed to OCR. Quality can be checked by assessing whether text strokes remain intact, background shadows are reduced, and detected text lines and regions align with the page layout. When blur is severe, image processing may improve readability only slightly; better focus, stable camera positioning, and adequate lighting are needed to capture missing detail.

Q3. Segmenting apples for defect detection

An apple-inspection system must distinguish each fruit from its surroundings and identify damaged areas despite shadows, reflections, variable brightness, and changing conveyor backgrounds. The segmentation method should be chosen according to the image conditions and the property being detected. A useful system often combines fruit localization with a separate analysis of surface defects rather than expecting one threshold to solve both tasks.

Global thresholding classifies pixels using one intensity or color cutoff for the whole image. It is simple and fast, and it can work when the apples and conveyor belt have consistently different colors and illumination is stable. For example, a fixed color range may isolate red or green fruit against a uniform background. Its main weakness is sensitivity to brightness changes. A shadow can make an apple fall below the selected threshold, while a reflection can make part of the apple appear similar to the background. Different apple varieties and background colors can also make a single cutoff unreliable.

Adaptive thresholding uses local image statistics to calculate a different cutoff in different neighborhoods. It can handle gradual lighting changes better than a global threshold and may help separate fruit from a background when one side of the conveyor is brighter than another. However, it is primarily based on local intensity, so strong texture, reflections, or small shadows can cause irregular masks. It may also divide a single apple into multiple regions if its surface brightness varies sharply. Adaptive thresholding can be useful when the illumination is uneven but should be combined with shape and color checks.

Edge-based segmentation identifies boundaries by locating strong changes in intensity or color. The curved outline of an apple may form a useful edge against the conveyor, and edge information can help refine an approximate fruit mask. Edge-based methods are less dependent on a single foreground brightness value, but they can fail when the fruit and background have similar colors or when shadows obscure part of the boundary. Reflections can create false internal edges, and a damaged region may have a weak boundary that is difficult to detect. Edge detection is therefore often more effective for locating fruit contours than for directly classifying defects.

Region-based segmentation groups neighboring pixels that have similar properties, such as color, brightness, or texture. Methods such as region growing or clustering can identify a fruit region even when its appearance varies gradually, provided there is a suitable starting point or similarity rule. Region-based techniques can adapt better than a single global threshold to naturally varying fruit color. Their limitations include sensitivity to the choice of seed, features, and stopping rules, as well as the possibility that reflections or shadows split the fruit into separate regions or cause regions to leak into the background.

A robust approach combines the methods. First, use a conveyor region of interest and a background model to reduce irrelevant pixels. Then identify candidate apple regions using color and intensity features, choosing global thresholding if lighting and background are stable, or adaptive thresholding and region-based grouping if illumination varies. Refine each candidate's outline with edge information and shape constraints, such as a compact, approximately rounded contour. If apples overlap, connected components alone may not separate them; a watershed-like separation, instance-segmentation model, or additional depth and shape cues may be needed.

After isolating an apple, defect detection should focus on its surface rather than the entire scene. The system can compensate for local illumination, then look for color changes, texture irregularities, spots, bruises, cuts, or regions with unusual edge patterns relative to the surrounding peel. Reflections should be treated cautiously: a bright highlight may resemble a pale defect, so a specular-reflection mask or color and texture checks can reduce false alarms. The final defect decision can combine region size, shape, color difference, texture, and position on the fruit. A practical sequence is: **normalize the image where needed; restrict processing to the conveyor area; segment each apple using color or adaptive/region-based methods; refine fruit boundaries with edges and shape; correct for shadows and reflections; segment candidate defect regions; and classify each fruit as acceptable or damaged**. The system should be evaluated across varieties, backgrounds, lighting conditions, and defect types, since a method that works on one uniform setup may not generalize to the production line.

Q4. Lane detection under changing road conditions

A lane-detection system processes a forward-facing camera image to find lane markings that guide a vehicle's position. Shadows, rain, faded paint, damaged markings, vehicles, and changing sunlight can all create visual patterns that resemble or obscure lane lines. The processing stages must therefore be designed as a sequence, with each stage improving one aspect of the image while recognizing its limitations. Image processing can support lane detection, but safety-critical vehicle systems also require temporal tracking, uncertainty

estimation, and sensor or model-based checks.

The first stage is **grayscale conversion**, which reduces a color image to a single intensity channel. This simplifies many filtering and edge-detection operations and can make the pipeline faster. It can also discard useful color cues: yellow lane markings, for example, may be easier to distinguish from a gray road in color than in grayscale. A system may therefore use grayscale for edge detection while retaining color information in a parallel lane-marking mask.

Filtering reduces noise and small visual variations before edge detection. A Gaussian filter can suppress fine noise and rain-related speckles, while a median filter may reduce isolated impulse-like artifacts. Strong filtering, however, blurs thin or faded lane markings, and large raindrops or motion blur cannot necessarily be corrected by simple smoothing. The filter size should be chosen to reduce irrelevant texture without removing the narrow edges needed to identify lane boundaries.

Contrast enhancement makes lane markings more distinguishable from the road. Contrast stretching or a local contrast method can help when markings are faint or when lighting varies across the image. Excessive local enhancement can make road texture, cracks, shadows, and reflections appear as strong features. Enhancement should therefore be limited and ideally tested across daytime, nighttime, tunnel, and wet-road footage. When contrast varies over the frame, local methods may help, but the system must avoid turning shadow boundaries into false lane edges.

Edge detection finds intensity transitions that may correspond to the boundaries of painted lines. A gradient-based detector can identify candidate edges, while further geometric reasoning can select edges that are long, suitably oriented, and consistent with the expected road perspective. Edge detection is vulnerable to shadows, road seams, vehicle outlines, guardrails, and reflections. Damaged or worn lane markings can produce discontinuous edges, so the system should not depend on one uninterrupted line.

Thresholding selects pixels likely to belong to lane markings based on brightness, color, or edge strength. A fixed threshold can be fast but may fail as illumination changes. Adaptive thresholds can respond to local brightness, while color thresholds can target common marking colors when color information is reliable. Wet roads can create bright reflections that pass a brightness threshold; shadows can make genuine markings too dark. Thresholding should therefore generate candidates, not final lane decisions, and should be combined with shape, location, and temporal consistency.

A **region of interest (ROI)** limits processing to the part of the image where lane markings are expected, usually the road surface ahead of the vehicle. This reduces false detections from the sky, buildings, signs, and roadside objects and saves computation. A fixed trapezoidal ROI may work on a level, straight road but can fail on hills, curves, vehicle pitch changes, or different camera installations. A flexible ROI based on road geometry or a perspective transformation can be more robust, but still requires validation when the vehicle turns or the road shape changes.

The overall sequence is: **capture and normalize the frame; preserve color information if useful; convert a processing copy to grayscale; filter carefully; enhance contrast; restrict analysis to the road ROI; detect candidate edges and threshold likely lane pixels; group candidates using their expected geometry; fit lane curves; and track the lane estimates across frames**. Temporal tracking can bridge short gaps caused by

damaged markings or temporary occlusion, but it should not blindly preserve a lane estimate after the road changes. Confidence should be reduced when rain, shadows, vehicles, or missing paint make the evidence weak. The system should then use additional cues or enter an appropriate safe behavior rather than treating an uncertain image-processing result as definitive.

Q5. Waste classification on a conveyor belt

A municipal waste-sorting system must classify objects such as plastic, paper, and metal while handling overlap, shadows, reflections, sensor noise, and changing orientation. A complete pipeline includes image acquisition, preprocessing, enhancement, object segmentation, feature extraction, classification, and quality control. The objective is not just to make an image look cleaner; each stage must preserve information that helps distinguish materials and must cope with the fact that real waste objects may be dirty, crushed, partly hidden, or visually similar.

The system first **captures and calibrates images** from a camera positioned above the conveyor. Consistent lighting can reduce shadows and reflections, while calibration can correct lens distortion and map image coordinates to the conveyor surface. If possible, the conveyor background should have a color and texture that differ from the waste. A stable camera, controlled exposure, and synchronized conveyor motion reduce blur. Additional sensors, such as depth or near-infrared imaging, can provide material cues that a normal RGB camera cannot reliably supply.

Preprocessing corrects image artifacts before object analysis. Denoising with a mild median or Gaussian filter can reduce sensor noise, but strong smoothing can remove printed text, thin edges, or surface texture. White balance and illumination normalization can reduce color shifts and gradual brightness variation. If the system uses video, motion blur and inconsistent exposure should be addressed at capture time where possible. Shadows and highlights should not simply be erased indiscriminately, because some variations carry useful information; the goal is to reduce nuisance effects while retaining material-specific cues.

Enhancement makes relevant details easier to measure. Contrast adjustment can improve visibility of boundaries, labels, folds, and surface texture. Local contrast enhancement may help with objects that have both bright and dark regions, but it can also amplify shadows and noise. Color normalization helps make categories less sensitive to changes in lighting, while a separate brightness or reflectance-related feature can help describe how a material responds to illumination. Reflective metal and glossy plastic can produce highlights, so brightness alone should not determine the class.

Segmentation separates the objects from the conveyor and from one another. Background subtraction is effective when the belt appearance is stable, while color or intensity thresholds can provide quick foreground candidates. Edge information can help locate boundaries, and connected-component analysis can identify separate regions. These approaches become difficult when objects overlap or touch. In such cases, shape-based separation, watershed methods, instance segmentation, or depth cues may be needed. Shadows should be distinguished from object regions by their softer boundaries, color properties, and position relative to objects; reflections should be treated as part of the object appearance but not mistaken for a separate object.

Once candidate objects are segmented, **feature extraction** describes each object in a form suitable for

classification. Useful features can include color statistics, shape, area, aspect ratio, contour complexity, texture, edge density, reflectance, and local patterns. Plastic may show smooth or glossy areas and a wide range of colors; paper may show fibrous texture, folds, print, and relatively matte surfaces; metal may exhibit strong highlights, distinctive reflectance, and characteristic shapes. No single feature is sufficient: a shiny plastic wrapper may resemble metal in brightness, and a printed or coated paper item may resemble plastic. Combining color, texture, shape, and reflectance features is more reliable than relying on color alone.

The **classification** stage assigns each object to a category such as plastic, paper, or metal. A trained machine-learning model can use the extracted features, or a convolutional neural network can learn features directly from labeled object images. Training data should represent the actual conveyor conditions: different material types, colors, orientations, degrees of damage, lighting, overlap, and background conditions. The system should be tested on images it did not use for training, including difficult cases such as crushed cans, transparent plastic, cardboard, wet materials, and partially hidden objects. If the classifier is uncertain, it should be able to reject the item for manual review or send it to a fallback sorting path rather than making an unsupported high-confidence decision.

The complete pipeline is: **capture and calibrate the image; denoise and normalize illumination and color; enhance relevant contrast; segment the conveyor foreground and individual objects; separate overlapping items using geometry, depth, or instance-level methods; extract color, shape, texture, and reflectance features; classify each object; and route uncertain or unclassified items for review**. Real-world performance should be monitored by class, not just as one overall accuracy figure, because a system may classify paper well but mistake reflective plastic for metal. Evaluation should include missed objects, incorrect class assignments, and performance under changing lighting and conveyor conditions. Periodic retraining with newly observed examples and regular camera and lighting checks help the system remain reliable as waste streams and operating conditions change.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date:	Date:	Date: